Transient processes in a complex circuit

W22 (2022) — English translation

The electrical circuit shown in the figure consists of three capacitors $C_1 = C_2 = C_3 = C$, two identical resistors $R_1 = R_2 = R$, a constant voltage source $\mathcal{E}$ with zero internal resistance and a switch K. While the key was open, the capacitors were not charged. Let us denote by t the time after closing the key, the charges of the capacitors by q_1, q_2 and q_3 and the current strength in them ($I = dq/dt$) I_1, I_2 and I_3 in accordance with their numbers.

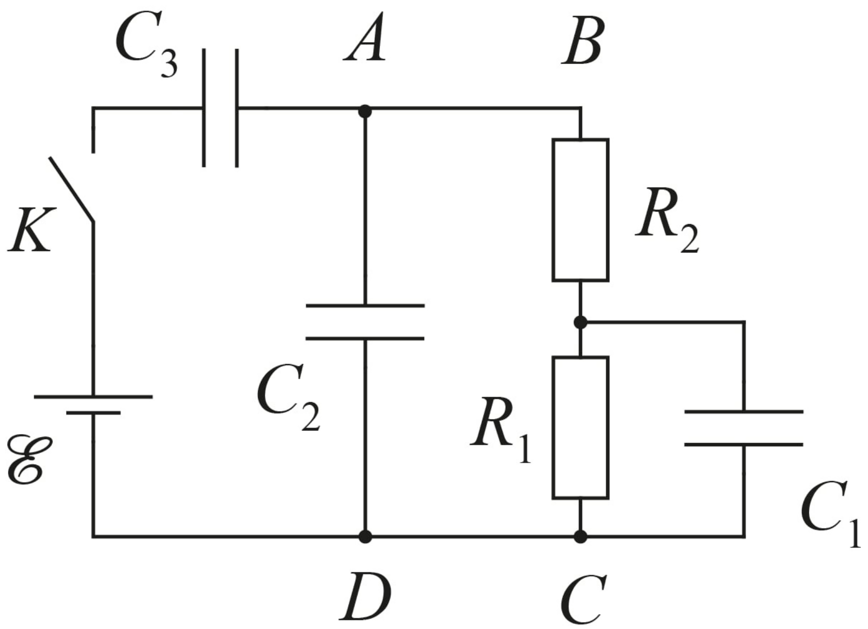

Figure 1: Figure 1.

A1	Find the charges of capacitors q_{1_0}, q_{2_0} and q_{3_0} at $t \to 0$.	**0.60**

A2	Find the charges of capacitors q_{1_∞}, q_{2_∞} and q_{3_∞} at $t \to \infty$.	**0.60**

A3	Find the amount of heat Q_0 released in the system during time $t \to 0$.	**0.40**

A4	Find the amount of heat Q_∞ released in the system during time $t \to \infty$.	**0.40**

Let's move on to finding the time dependence of the charge of the first capacitor $q_1(t)$.

A5	From Kirchhoff's first law written for node A, express I_2 in terms of I_1, q_1, R and C.	**0.50**

A6 From Kirchhoff's second law written for contour $ABCD$, express q_2 in terms of I_1, q_1, R and C. Also find I_2 by differentiating q_2 with respect to time. Express your answer using $\frac{dI_1}{dt}$, I_1, R and C. **0.50**

A7 Equating the expressions for I_2 obtained in paragraphs A5 and A6, show that the differential equation describing the dependence $q_1(t)$ has the following form

$$nalphanfracd^2 q_1 dt^2 + nbetanfracdq_1 dt + q_1 = 0.$$

α and β R and C. **0.40**

Further, consider it known that the solution to this differential equation must be sought in the form

$$q_1(t) = Ae^{nlambdat},$$

where λ from R and C, A – from the initial conditions.

A8 Find λ_1 and λ_2 that satisfy the equation obtained in A7. Express your answers using R and C. For λ_1, select the higher of the values. **0.50**

In fact, the dependence $q_1(t)$ is represented as the sum of the found solutions

$$q_1(t) = A_1 e^{nlambda_1 t} + A_2 e^{nlambda_2 t}$$

A9 From the initial conditions, find A_1 and A_2. Express your answers using C and $\mathcal{E}$. **0.80**

A10 Find the amounts of heat Q_{R_1} and Q_{R_2} released in the resistors during time $t \to \infty$. Express your answers using C and $\mathcal{E}$. **1.30**

: $\int e^{ax} dx = \frac{e^{ax}}{a} + C$

Source: `https://pho.rs/p/274`